

Sak 19.3.3

Solutions

$$V(r) = V_0 e^{-r^2/r_0^2}$$

$$\textcircled{1} f(\theta) = -\frac{2\mu}{\hbar^2} \int_0^\infty \frac{\sin qr}{q} V_0 e^{-r^2/r_0^2} r dr \quad q = 2K \sin(\theta/2)$$

$$= -\frac{\mu V_0}{\hbar^2 q} \left[\int_0^\infty \left(\frac{e^{iqr} - e^{-iqr}}{i} \right) e^{-(r/r_0)^2} r dr \right]$$

$$= -\frac{\mu V_0}{\hbar^2 q} \exp\left(-\frac{r_0^2 q^2}{4}\right) \left[\int_0^\infty u e^{-u^2/r_0^2} du + \int_0^\infty \frac{r_0^2 i q}{4} e^{-(u/r_0)^2} du \right. \\ \left. - \int_0^\infty u e^{-(u/r_0)^2} du + \int_0^\infty \frac{r_0^2 i q}{4} e^{-(u/r_0)^2} du \right]$$

where $u \equiv r + \frac{r_0^2 i q}{r}$

$$\textcircled{2} f(\theta) = -\frac{\mu V_0}{i \hbar^2 q} r_0^2 i q \sqrt{\pi} r_0 e^{-r_0^2 q^2/4}$$

$$|f(\theta)| = -\frac{\mu V_0 r_0^3 \pi}{2 \hbar^2} \exp\left(-\frac{r_0^2 q^2}{4}\right)$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \frac{\mu^2 V_0^2 r_0^6 \pi}{4 \hbar^4} e^{-r_0^2 q^2/2}$$

$$\textcircled{3} = \iint \frac{\pi r_0^2}{4} \left(\frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 \exp\left(-\frac{q^2 r_0^2}{2}\right) \sin\theta d\theta d\phi$$

$$= \frac{\pi r_0^2}{4} \left(\frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 \iint e^{-\frac{q^2 r_0^2}{2}} \sin\theta d\theta d\phi \quad q^2 = 2K^2(1-\cos\theta)$$

$$\textcircled{4} = \frac{\pi r_0^2}{4} \left(\frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 2\pi \int_0^\pi e^{-\frac{r_0^2}{2} \cdot 2K^2(1-\cos\theta)} \sin\theta d\theta$$

let $u = (1-\cos\theta) \quad du = \sin\theta d\theta$

$$= \frac{\pi^2 \Gamma_0^2}{2} \left(\frac{\mu V_0 \Gamma_0^2}{\hbar^2} \right)^2 \int e^{-\Gamma_0^2 k^2 u} du$$

$$= \frac{\pi^2}{2} \left(\frac{\mu V_0 \Gamma_0^2}{\hbar^2} \right)^2 (1 - e^{-\Gamma_0^2 k^2 \cdot 2})$$

$$\sigma = \frac{\pi^2}{2k^2} \left(\frac{\mu V_0 \Gamma_0^2}{\hbar^2} \right)^2 (1 - e^{-\Gamma_0^2 k^2 \cdot 2})$$

$$(2) \quad \mathcal{H} = \int d^3k \sum_{\lambda} \left(\frac{1}{2} P_{\lambda k}^2 + \frac{\omega^2}{2} Q_{\lambda k}^2 \right)$$

So for any particular λ, k mode, take functional form:

$$P_{\lambda k} = i\omega_k Q_{\lambda k}^0 e^{i\omega_k t} \quad Q_{\lambda k} = Q_{\lambda k}^0 e^{i\omega_k t}$$

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial P_{\lambda k}} &= P_{\lambda k} \quad , \quad \frac{\partial \mathcal{H}}{\partial Q_{\lambda k}} = \omega_k^2 Q_{\lambda k} \\ &= i\omega_k Q_{\lambda k}^0 e^{i\omega_k t} \quad = \omega_k^2 Q_{\lambda k}^0 e^{i\omega_k t} \\ &= \frac{dQ_{\lambda k}}{dt} \quad = - \frac{dP_{\lambda k}}{dt} \end{aligned}$$

--- which are Hamilton's equations.